Tug Allocation QAOA on IBM Hardware: 60-Run Technical Note

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Summary

Across 59 runs of one fixed 18-qubit QAOA circuit on ibm_kingston, 0.774% ± 0.142% of shots (mean ± sample s.d.; s.e.m. 0.019%) were the proven-optimal tug schedule, and all 59 runs sampled it at least once. That rate is about 85 times uniform random sampling (0.0092%) and about 22% of the noiseless simulator (3.44%). The run-to-run spread matches binomial shot noise at 4,000 shots (dispersion test p = 0.37). An earlier unoptimised compilation on ibm_fez (run 1) gave 0.30% (z = 3.4, p ≈ 0.0007). Total IBM time for all 60 runs was 96 s.

Problem and circuit

The instance assigns 4 tugs to 4 ships (2 contract, 2 spot) at 12 kn wind, with demands of 1, 0, 1 and 1 tugs. Binary variables x(t,s) (tug t serves ship s, 16 variables) and y(s) (spot ship s accepted, 2 variables) give 18 qubits. The QUBO combines tug cost, minus spot revenue, plus quadratic penalties for double-booking a tug, missing a contract demand and serving a spot ship off its demand. It has 18 linear and 56 quadratic terms. Exhaustive search over 2^18 = 262,144 states gives the optimum energy −9,000, reached by 24 degenerate states; 52 states satisfy every constraint.

The ansatz is QAOA with p = 1 and angles (γ, β) = (0.7581, 2.6581). Compilation used Qiskit optimization level 3 with transpiler seed 172. That seed and the machine were chosen from 300 seeds on each of 3 machines by estimated success probability, ESP = Π(1 − ε) over all gates and measurements from live calibration. The compiled circuit has 204 CZ gates, depth 316 and ESP 0.546. Run 1 used seed 1 on ibm_fez: 237 CZ, ESP 0.408.

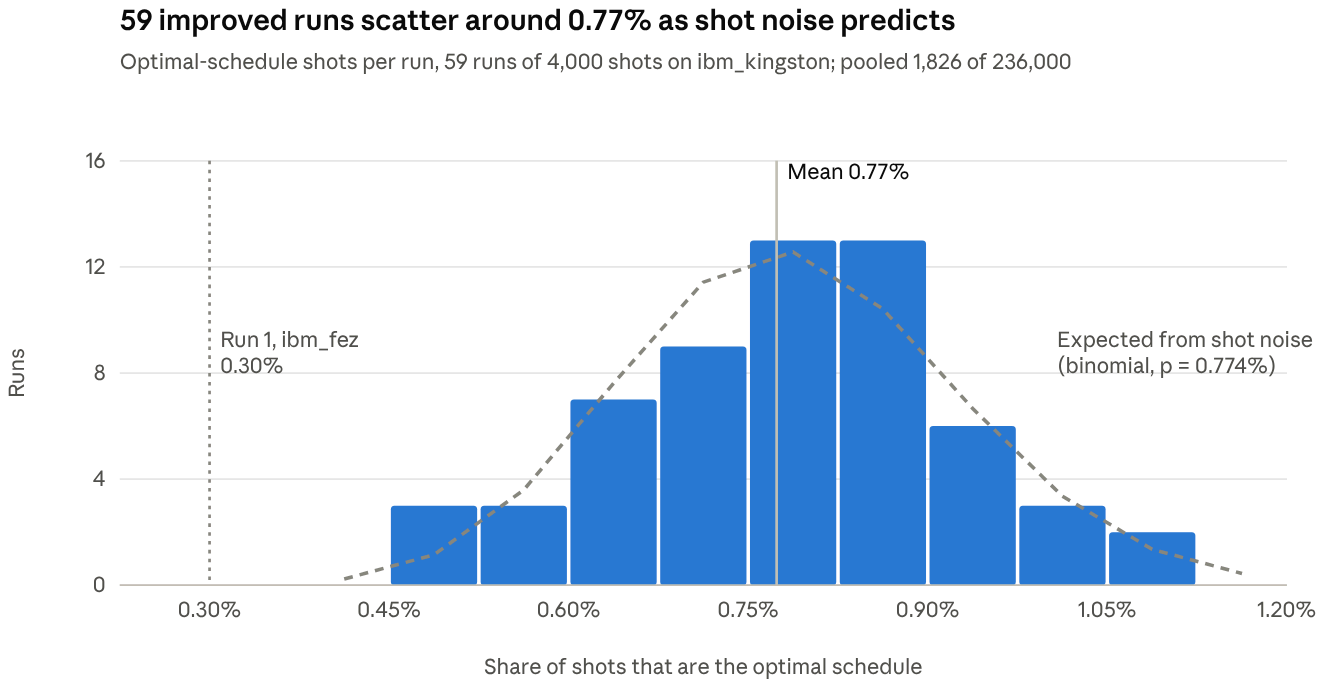
Protocol

Every run is 4,000 shots through SamplerV2 with dynamical decoupling and gate and measurement twirling, under a 180 s execution cap per job. Runs 2 to 4 (00:05 to 00:19 UTC, 18 Oct 2026) were one job each. Runs 5 to 60 (01:03 to 01:05 UTC) reused the

identical compiled circuit, with the same 18 physical qubits, as one PUB each. They were packed 8 PUBs per job into 7 jobs in one Batch, a pilot job first, which halved the cost to 1.5 s per run. Twirling instances are drawn independently per PUB.

A shot is *optimal* if its QUBO energy equals −9,000 and *valid* if it breaks no constraint. All runs are scored by the same function (`score_counts`). No error mitigation beyond twirling and decoupling, and no post-selection, is applied to the reported shares.

Results



results/hardware/*/run.json and runs.csv in fdumary/Tug-Boat-Optimizer · runs 1 to 60, 18 Oct 2026

Quantity	Value
Optimal share, runs 2–60 (n = 59)	0.774% ± 0.142% s.d.; s.e.m. 0.019%; median 0.775%; range 0.450–1.075%
Valid share	1.70% ± 0.20%
Optimal among valid	45.6% ± 6.0% (noiseless simulator 47.8%)
Runs sampling the optimum	59 of 59 (60 of 60 with run 1)
Dispersion vs one binomial rate	χ^2 = 61.1, 58 d.o.f., p = 0.37; expected s.d. 0.139%
Between the 7 jobs of runs 5–60	one-way ANOVA F = 0.32, p = 0.92

Quantity	Value
Runs 2–4 vs runs 5–60 (pooled)	0.692% vs 0.778%; z = 1.05, p = 0.29
Runs 2–60 vs run 1 (pooled)	1,826/236,000 vs 12/4,000; z = 3.4, p ≈ 0.0007
Ratio to uniform sampling (24/2^18)	≈ 85×
Ratio to noiseless simulator (3.44%)	≈ 0.22

The dispersion test compares per-run optimal counts against a single shared binomial rate. The other comparisons are pooled two-proportion z-tests. Hardware loses most of the simulator's signal to rule-breaking shots: the valid share falls from about 7.4% to 1.7%. Among valid shots, the optimal fraction stays close to the simulator's.

Limitations

- **One calibration window.** Runs 5 to 60 ran within about 2 minutes, and runs 2 to 60 within about 1 hour. The data bound shot-to-shot and run-to-run variance, not drift across days or recalibrations.
- **Confounded comparison with run 1.** Run 1 differs in both machine and compilation. The ESP estimates (0.408 to 0.447 on ibm_fez from the seed, 0.546 on ibm_kingston) suggest most of the gain comes from the machine.
- **One instance, one depth.** The results cover 4 tugs at 12 kn with p = 1 only, so they say nothing about scaling. Classical exhaustive search solves this 2^18 instance in seconds.
- **Shared jobs.** PUBs in the same job are not independent jobs, although the ANOVA finds no job effect (p = 0.92).
- **Noise model not validated.** The ESP-based model used to choose the machine and seed overpredicted the optimal share by 2 to 3 times. It was used only to rank options, never to report results.

Reproducibility

All code and raw counts are in fdumary/Tug-Boat-Optimizer on main, under `results/hardware/`. Each run folder has `run.json` and raw counts, and the repeats folder has `runs.csv`, `batch.json`, `stats.json` and `counts/run_NN.json`.

`scripts/summarize_hardware_runs.py` recomputes every statistic above from the saved counts, with no credentials and no QPU time. The software was Qiskit 2.6.0, qiskit-ibm-runtime 0.50.0 and Python 3.13.

Runs	Script	IBM job IDs
1	`run_hardware_experiment.py`	dba0j7vi0auc7391b3tg (ibm_fez)
2–4	`run_hardware_improved.py`	dba0ql8c77vs738253fg, dba0veni0auc7391blug, dba1168c77vs73825bsg
5–60	`run_hardware_repeats.py`	dba1lmhgv2cc73e67ucg, dba1m1424mrc7384ou10, dba1m19gv2cc73e67v3g, dba1m19gv2cc73e67v40, dba1m1k24mrc7384ou2g, dba1m1gc77vs73826dug, dba1m1s24mrc7384ou4g (8 runs each, in order)

`run_hardware_repeats.py --plan` recompiles the circuit and checks it against run 2 (CZ count and physical qubits) without submitting. It refuses to run if calibration has changed the compilation.